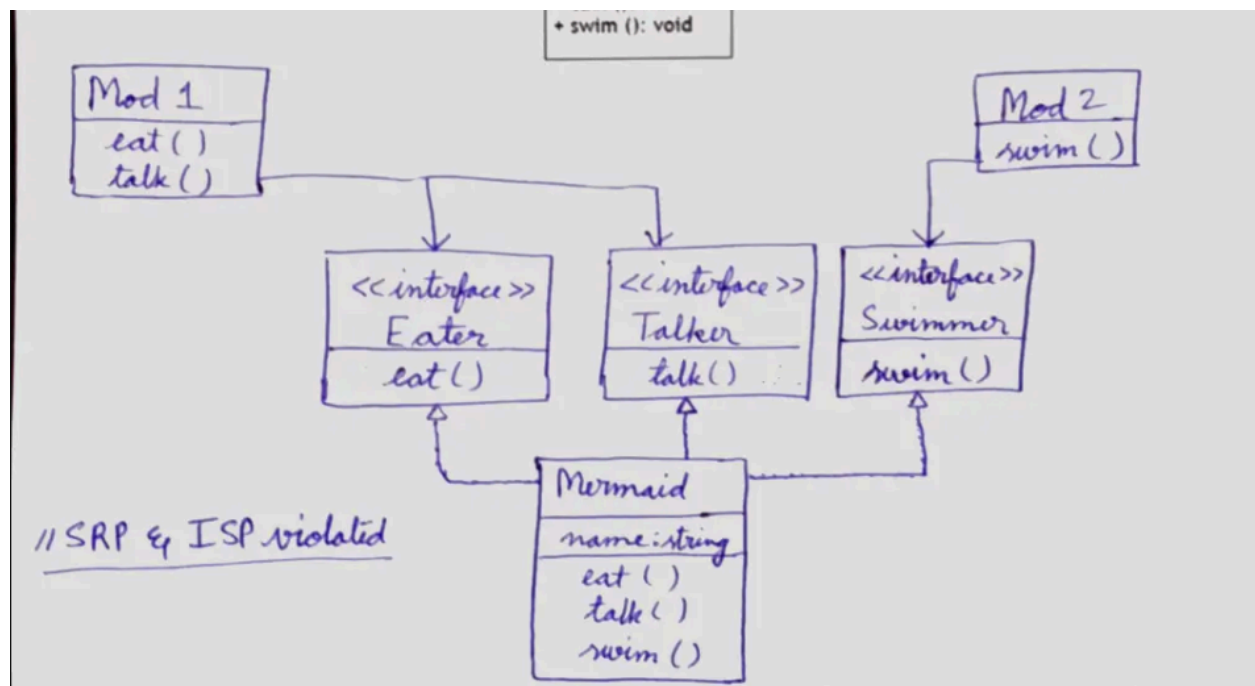


Quiz 4 - Version 1 Solution:

Q1(a) : Consider a video game that has different creatures and characters. One interesting creature is a mermaid . The mermaid eats and talks like a woman , and swims like a fish. The mermaid class provides all three functions. This game has several modules. One of these modules (Mod 1) uses only the eat and talk functions of the mermaid while (Mod 2) uses only the swim function. Refactor (improve) the design below in the light of solid principles . Also mention the principles that are violated in this design.



(b) ISP -> Interface Segregation Principle

(c) OCP-> Open Closed Principle

Object Oriented Metrics Solution :

Sr#	Class	Metrices	Value
1	Authority	Weighted Methods per Class	2
2	License	Weighted Methods per Class	1

3	HTV	Weighted Methods per Class	3
4	Vehicle	Depth Of Inheritance(DIT)	0
5	LTV	Depth Of Inheritance(DIT)	1
6	HTV	Depth Of Inheritance(DIT)	1
7	Vehicle	Number Of Children (NOC)	2
8	LTV	Number Of Children (NOC)	0
9	Person	Coupling between objects (CBO)	3
10	Vehicle	Coupling between objects (CBO)	3